HW6: Dimensionality Reduction: PCA, Kernel PCA, t-SNE, and UMAP

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This report compares four dimensionality reduction methods on the Fashion-MNIST subset: PCA, Kernel PCA with an RBF kernel, t-SNE, and UMAP. The dataset contains 2000 observations sampled in a class-balanced way from the original data, and every image is represented in R^{784}. All methods were applied to standardized data, and for t-SNE and UMAP I tried multiple hyperparameter settings and selected the final setting by visual inspection of cluster separation, overlap, and overall structure.

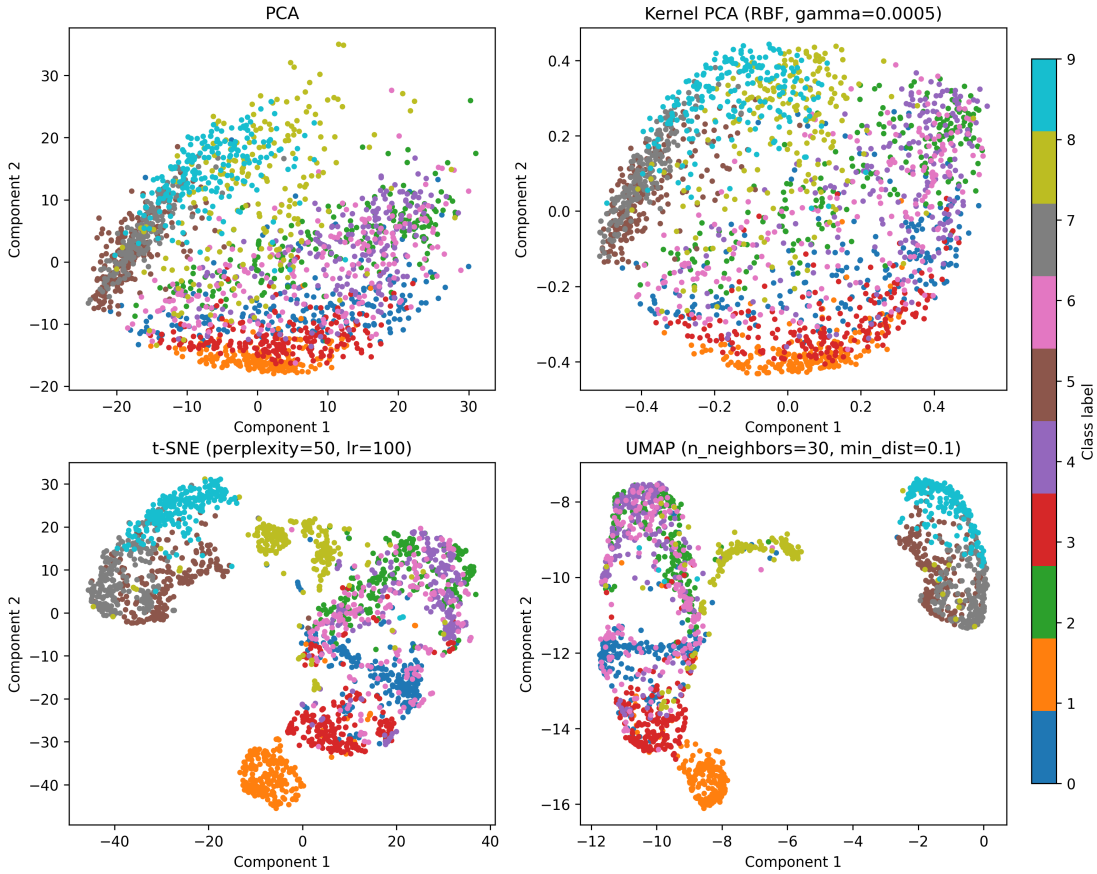

Figure 1: Final comparison of the four methods. The selected nonlinear settings are t-SNE with perplexity = 50 and learning rate = 100, and UMAP with $n_neighbors = 30$ and $min_dist = 0.1$.

PCA. The PCA visualization shows a massive central cluster where most data points heavily overlap. Although colors like cyan and orange pull slightly toward the outer edges, the linear projection cannot effectively resolve the distinct categories, making it difficult to distinguish the individual classes from one another.

Kernel PCA. Applying Kernel PCA with an RBF kernel and a gamma of 0.0005 forces the data into a more circular overall arrangement. Despite this change in global shape, the internal

mixing of classes remains just as problematic as in standard PCA. The overlapping data points indicate that this nonlinear transformation is not enough to pull the groups apart.

t-SNE. Configured with a perplexity of 50 and a learning rate of 100, t-SNE does an excellent job of forming distinct, localized islands. The orange category is completely isolated at the bottom, and other colors form tight, easily identifiable clusters. There is still some intermingling on the right side of the plot, but the local separations are vastly improved.

UMAP. Using 30 neighbors and a minimum distance of 0.1, UMAP organizes the points into highly dense, almost solid structures. The layout is broken into a few major groupings separated by large empty spaces. This approach highlights strong global connectivity among certain subsets of classes, though the tight packing makes the internal boundaries of those larger islands slightly harder to parse than in the t-SNE projection.

Runtime comparison. To compare computational cost, I measured the full pipeline used for each final method over three runs and report the mean time in Table 1. The nonlinear methods are much slower than PCA-based methods, especially after including the PCA-50 preprocessing step used before t-SNE and UMAP.

Method	Mean runtime (s)	Relative to PCA
PCA	0.048	1.00
Kernel PCA	0.175	3.65
t-SNE	6.671	139.03
UMAP	17.764	370.22

Table 1: Runtime comparison for the final selected settings.

So the relative runtime comparison is approximately

$$\text{PCA} : \text{Kernel PCA} : \text{t-SNE} : \text{UMAP} = 1 : 3.65 : 139.03 : 370.22.$$

PCA is clearly the fastest method, followed by Kernel PCA. Among the nonlinear methods, t-SNE is much slower than PCA but still substantially faster than the chosen UMAP configuration in this experiment.

Conclusion. If the primary objective is to achieve clear visual separation of the different categories, I would select t-SNE with a perplexity of 50 and a learning rate of 100. It successfully breaks the data down into distinct, identifiable islands, making it the most effective tool for local clustering among the options tested. Both standard PCA and Kernel PCA struggle significantly with overlapping data points, making them mostly unhelpful for resolving individual classes. UMAP with 30 neighbors and a minimum distance of 0.1 offers an interesting alternative by showing strong global connectivity across large structures, but its extremely dense packing obscures the finer internal boundaries. For the most readable and cleanly separated visual clusters, t-SNE clearly stands out as the best approach in this comparison.